

Artificial Intelligence & Machine Learning – Task 2

Feature Engineering, Model Optimization & Performance Comparison

1. Objective

Apply feature engineering and feature scaling, train multiple regression models on the California Housing dataset, and compare their performance using RMSE and R^2 .

2. Dataset

The California Housing dataset was used for house-price prediction. It contains eight input features and a continuous target representing median house value.

3. Preprocessing

The target was separated from the input features. StandardScaler was used for feature scaling. The data was divided into training and testing sets using an 80:20 split with `random_state=42`.

4. Models Implemented

Model	Description
Linear Regression	Baseline linear regression model
Ridge Regression	Linear regression with L2 regularization
Decision Tree Regressor	Tree-based regression model

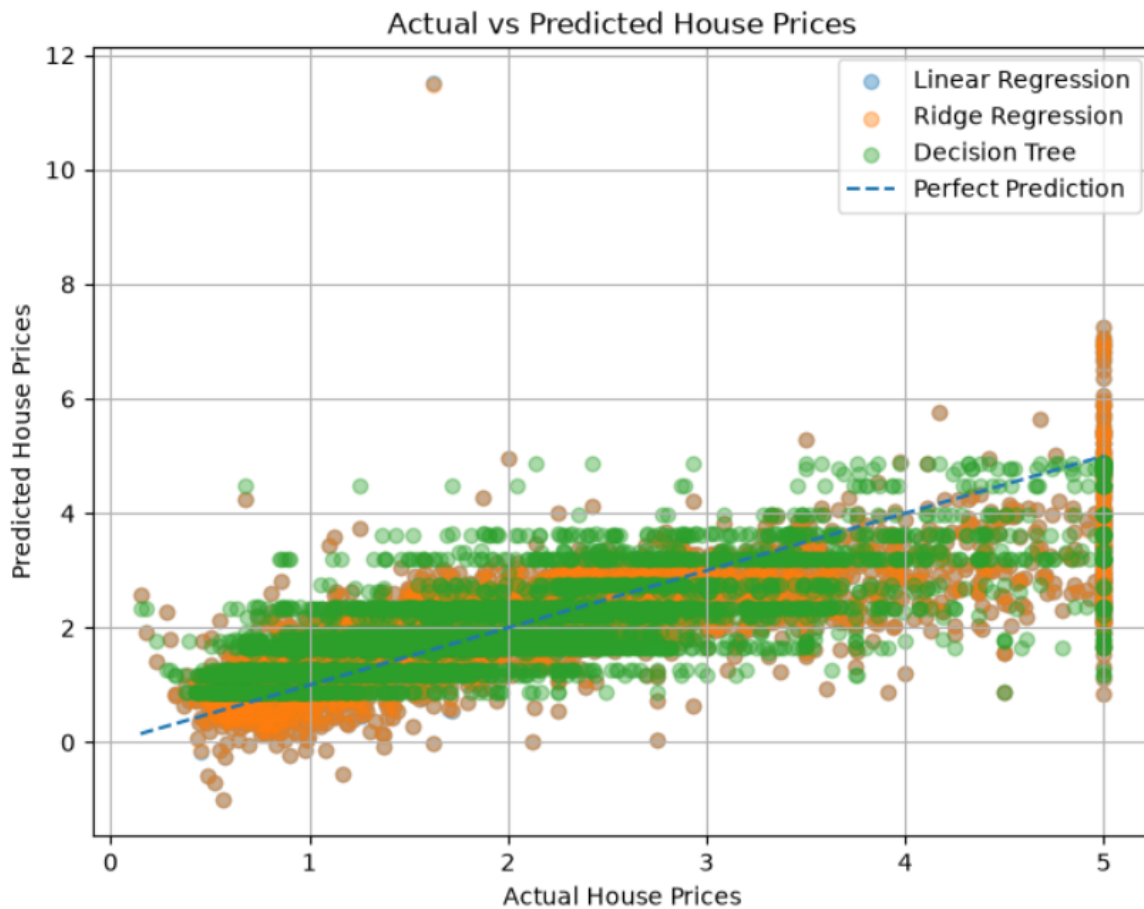
5. Performance Comparison

Model	RMSE	R^2
Decision Tree	0.724234	0.599732
Ridge Regression	0.745554	0.575819
Linear Regression	0.745581	0.575788

RMSE: lower values indicate smaller prediction errors. R^2 : higher values indicate greater explained variation. On this test split, the Decision Tree produced the lowest RMSE and highest R^2 among the evaluated models.

6. Actual vs Predicted Visualization

The plot compares actual house prices with predictions from the three trained models. The dashed line represents perfect prediction.



7. Conclusion

Three regression models were implemented and evaluated on the California Housing dataset. The experiment included feature scaling, an 80:20 train-test split, model training, and evaluation using RMSE and R^2 . The Decision Tree Regressor obtained an RMSE of 0.724234 and an R^2 score of 0.599732 on the test dataset. Ridge Regression and Linear Regression produced very similar results.

8. Tools Used

Python, pandas, NumPy, Matplotlib, scikit-learn, and Jupyter Notebook.